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# Fossil Fuel Prices and Capital Cost: A Machine Learning Driven Study on Energy Transition

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**An AI workflow connects energy-price signals with financing scenarios.**
**Regional fit differs, wind and solar fit most closely, and time-split checks show where price signals add information beyond a trend.**

## 01 Question and data

How do fossil fuel prices and inflation relate to energy-project WACC across China, Europe, and the Middle East?

**3 regions 9 technologies 120 monthly rows**

Historical price and macroeconomic inputs are paired with 2026–2035 WACC projections from Calcaterra et al. (2024). Coal and gas generation are conventional-technology comparisons.

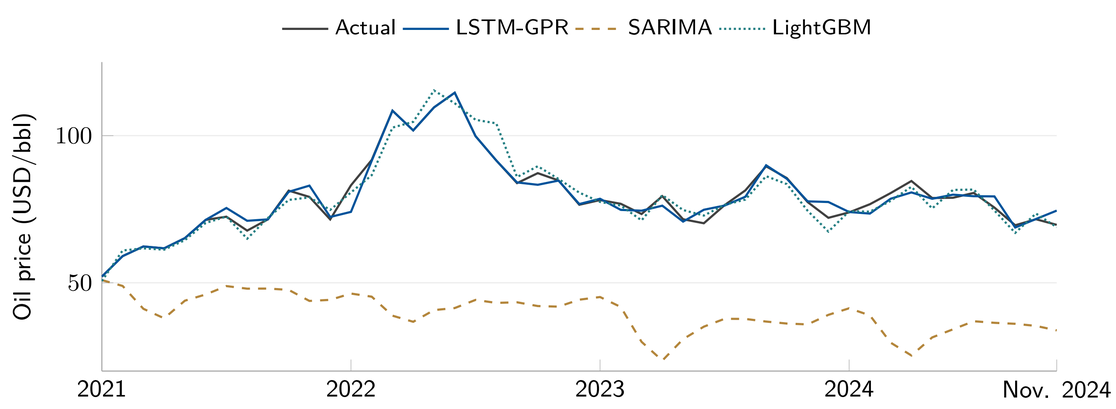
## 02 Three-stage framework

**1 Fuel-price forecasts**
CNN, LSTM/GRU, attention, and Gaussian-process smoothing

**2 Regional inflation**
LSTM + feed-forward network; five-model ensemble

**3 WACC comparisons**
AutoGluon regressions by region, fuel, and technology

## 03 Archived price comparison




LSTM-GPR: $R^2 = 0.963$, MAPE 1.86%, RMSE 2.416 USD/bbl. Metrics were recomputed from 47 saved monthly comparison outputs; this is separate from the fixed-origin baseline exercise.

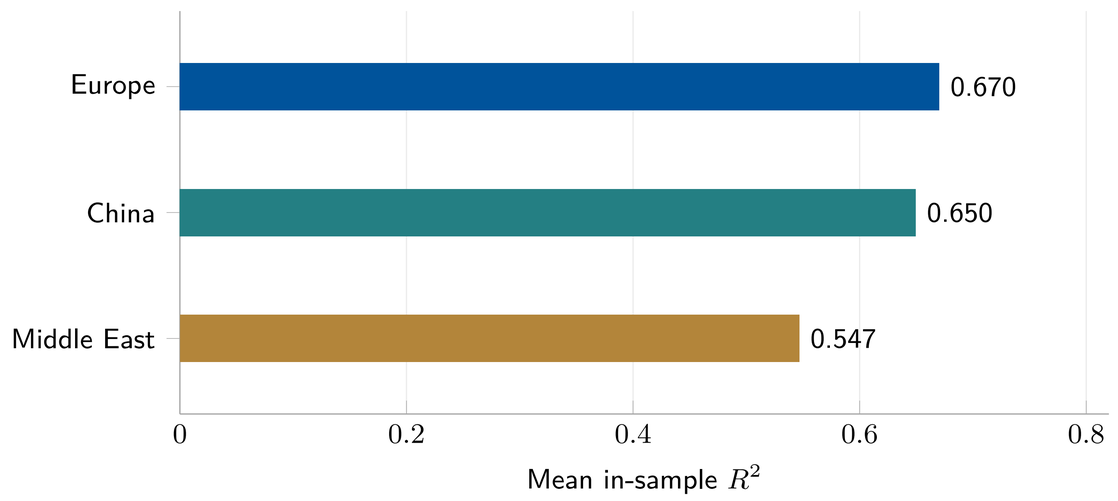
## 04 Open science and contact

Code + materials

Dataset snapshot

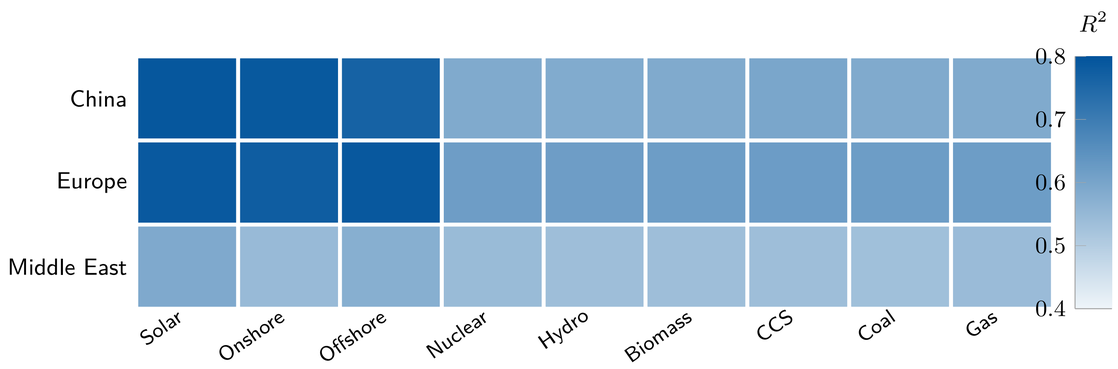
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Editable LaTeX sources and corrected diagnostics are in the GitHub repository.

## 05 Regional model fit



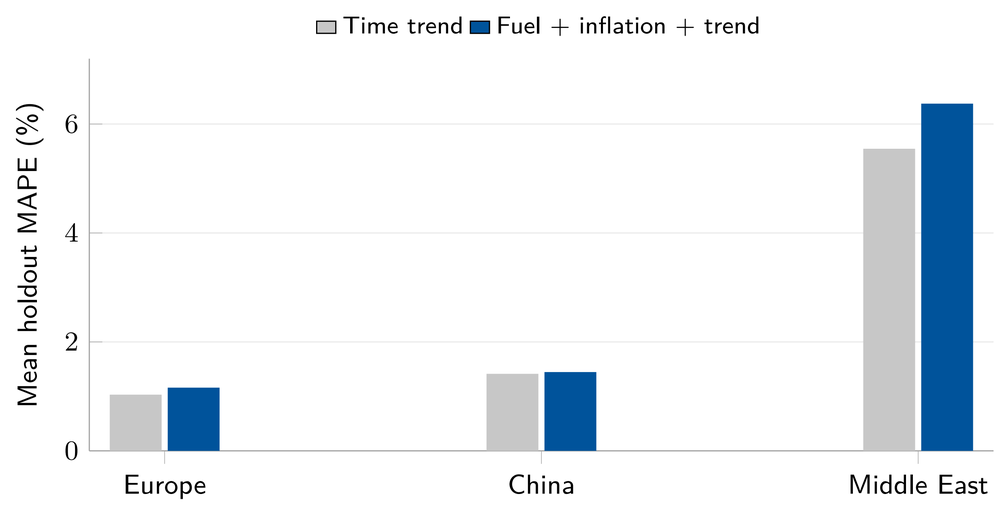

Europe and China have stronger original in-sample fit. These $R^2$ values describe the joint fuel-price and inflation regressions; they do not measure price-shock magnitude.

## 06 Technology differences




Mean $R^2$: solar 0.718; offshore wind 0.706; onshore wind 0.700. Other technologies average around 0.58. Cells average across fuels.

## 07 Chronological checks




Ridge holdout, 2033–2035. Fuel + inflation + trend improves out-of-sample $R^2$ over a time trend in China 8/27, Europe 6/27, and Middle East 6/27 cases.

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Selected references

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Calcaterra et al., *Nature Energy* 9, 1241–1251 (2024). doi:10.1038/s41560-024-01606-7.
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OpenAI Codex assisted language and consistency revisions throughout the manuscript, preparation of Fig. 1, and correction of the sensitivity-analysis code.